

disruptors—electric vehicles, autonomous vehicles, and the connected vehicles—ride on top of a very disruptive business model.

How do we refocus our R&D to enable the vision of sustainable mobility? What are those areas where we should refocus our energies?

There are three aspects to this. One is, what is the objective of automotive R&D, second, what are major shifts revolving around this, and third, what should be the strategy to respond to this shift. The primary objective should be to build a pipeline of technologies that improve product performance and deliver value to the customer. Another would be to explore disruptive technologies and assess their feasibility and suitability to adopt, and to build a strong portfolio of patents that enable commercialisation of the technology.

It is clear when you look at the current internal combustion engines that run on fossil fuels, that there is a saturation of this technology. It is time for us to look at what's next. This is where EVs come in because of zero emission, higher efficiency, and other attractive features, which makes them a very promising option for replacing the current engine technology.

When this happens, we also need to remember that the performance of the end product, which could be electric, autonomous and/or connected vehicles, is not that of the product alone. It is the product and service together which delivers the end performance to the customer.

There will be certain components that will not be needed in an EV such as certain engine parts, clutch, radia-

Challenges in adoption of EVs in India

There are always two angles to it. What is happening globally and what is happening in India, so that we can realise what needs to be done. To do that, one needs to understand the challenges.

The first one is range anxiety. We are always anxious about the limited energy carried by the battery. The next big issue is the cost of the battery itself, which is almost 50% of the cost of an EV. Third is the lack of charging infrastructure in India at present, especially fast charging.

Another bottleneck is the lack of availability of battery materials like lithium and cobalt, and rare earth for magnetic motors.

A recent report by Grant Thornton Bharat-FICCI said that India needs about 400,000 charging stations to meet the requirement for two million EVs that could potentially ply on its roads by 2026.

tor, and gears. There are also some parts that will continue in EVs such as steering system, seats, brake linings, headlights, and shock absorbers.

There is a whole new range of suppliers that will be created for electric motors, batteries, microprocessors, controllers, etc.

Open sourcing of technology

The technologies that are enabling electric vehicles, autonomous and the connected vehicles are growing exponentially, and their changes are non-linear. Multiple disruptive technologies are converging to co-create a higher level of performance. It is also a business model disruption.

There is a new trend emerging of open sourcing of technology. When it comes to IP creation itself, new players and practices are emerging. In the past, IP was created by the auto industry on its own, now it is a combination of auto and tech professionals who come together to understand and tackle the problem together to create solutions.

We have to visualise future scenarios and create technology road-

Recycle and reuse

The prices of lithium and cobalt have been quite volatile, and we need to have a better grip on the demand and supply of these materials. These materials are available only in certain parts of the globe. So, we need to mitigate the risks around the supply of these key battery materials.

We can look at deep sea mining of the critical materials from the oceans. But besides mining of these materials, we should design and use battery packs in such a way that these can be reused and recycled easily.

If we do all of this correctly, we will be able to create a sustainable model of mobility for the future. Today it is all about the ability to reimagine, refocus, reskill, rebuild, and reuse. The future of mobility lies in coherently working on these factors.

Reskilling is the way to go

The future mobility engineers need to have strong engineering fundamentals spanning across three or four divisions of engineering. Though mechanical engineering is still at the heart of it, on either side is electrical engineering and computer science. A mobility engineer in 2030 would need to be equally good in all three domains.

On top of it, a mobility engineer would need integration skills like quality management, systems management, communication, and customer orientation. He should also come with an innovative mindset, ethics, management skills, and be an avid learner. **EFY**



Autonomous car and self-driving concept

maps and align our efforts to these roadmaps. We also have to systematically evaluate the disruptive potential of new technologies and collaborate widely to have open innovation approaches. We should have the right IP approach to promote an encouraging environment.

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